# BRIAN JASON TRAN

Synthetic Data, Simulation, & Procedural Systems Lead
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## PROFESSIONAL SUMMARY

Synthetic data, simulation, and procedural systems lead with 10+ years of experience building scalable 3D pipelines for computer vision, robotics, generative AI, and media production. Specializes in Houdini and Python workflows for procedural content generation, fluid and rigid-body simulation, volumetric effects, USD authoring, and synthetic data pipelines. Combines deep 3D technical expertise with cross-functional leadership to develop scalable generation systems, improve data quality, and accelerate ML model development.

## EXPERIENCE

**Apple via The Select Group** | Seattle, WA (Hybrid) — Oct 2025 – Present
Technical Artist | Procedural & Agentic Workflows
- Built procedural scene-generation pipelines using Houdini, USD, and Python to produce tens of thousands of scene variations while minimizing redundant samples.
- Developed procedural modeling, material, and automation workflows to scale photorealistic 3D asset production and reduce manual iteration.
- Coordinated external artists and established quality-control standards for photorealistic deliverables.

**Irreverent Labs** - Backed by Andreessen Horowitz (a16z) | Bellevue, WA (Remote) — Mar 2023 – Oct 2024
Principal Technical Artist | Synthetic Data for Generative AI
- Built the Synthetics Division from a solo role to an 8-person team, delivering datasets for generative video and 3D model training.
- Set technical direction with ML engineers and executive leadership, defining project scope and selecting tools to accelerate model iteration.
- Led end-to-end delivery across internal teams and external contractors, balancing technical quality, timelines, budgets, and resources.

**Scale AI** | San Francisco, CA (Remote) — May 2022 – Feb 2023
Lead Technical Artist | Synthetic Data for Computer Vision
- Led synthetic data strategy across 5+ concurrent client projects spanning classification, robotics, OCR, and keypoint detection.
- Scaled a client's training dataset by 30×, with model IoU improving from 0.657 to 0.794.
- Developed Houdini/Python data-generation pipelines, managed 15+ distributed contractors, and drove improvements in labeling accuracy and quality control.

**Synthesis AI** | San Francisco, CA (Remote) — Dec 2020 – May 2022
Lead Technical Artist | Synthetic Data for Computer Vision
- Designed procedural generation systems for realistic human and environment datasets, supporting contract expansions with Apple, Amazon, and Google.
- Built parameterized environment and object workflows balancing photorealism with ML-engineer control; established technical standards across internal artists and external contractors.

**Walt Disney Animation Studios —** Simulation Technical Artist | Burbank, CA — May 2018 – Sept 2021
- Developed and executed simulation-driven effects workflows spanning fluids, volumetric phenomena, particles, and rigid-body dynamics for feature animation.

**Freelance —** Simulation Technical Artist | Los Angeles, CA / New York, NY / London, UK — 2016 – 2020
Delivered simulation-driven visual effects and FX supervision for projects involving Snapchat, Taylor Swift, and Maroon 5.

**Industrial Light & Magic —** Rendering and Pipeline Engineer | San Francisco, CA — Jan 2013 – May 2015
Provided technical and pipeline support for feature films.

## EDUCATION

**Bournemouth University** | National Centre for Computer Animation – Poole, United Kingdom
Master of Science, Computer Animation & Visual Effects (Graduated with Distinction)

**University of Illinois at Urbana-Champaign** | College of Engineering – Champaign, IL
Bachelor of Science, Computer Science (Specialization in Computer Graphics)

## SKILLS

- **Procedural & 3D:** Houdini, Houdini TOPs, USD, VEX, Python, Maya, V-Ray, Unity, Unreal Engine
- **Synthetic Data & ML:** Annotation, metadata integration, domain-gap analysis, t-SNE, UMAP, data validation, testing harnesses
- **Technical Leadership:** Cross-functional delivery, contractor and vendor management, QA/QC, resource allocation, budgeting